

exponentially. Quantum algorithms are sets of instructions that tell quantum computers what to do to solve problems.

The algorithm would, in particular, greatly speed up semidefinite programs that are used to learn unknown quantum states. Fernando Brandão, a professor in theoretical physics, said that this type of quantum learning problem is faced by researchers who study large quantum systems in a variety of different systems such as superconducting qubits, which are quantum information units similar to computer bits that would operate based on superconducting technology. The semidefinite programs are used to give a description of how the quantum matter is behaving, which, in turn, allows researchers to better understand the bizarre states of the subatomic world.

Smart windows that turn dark within a minute

Stanford University engineers have developed dynamic windows that can switch from transparent to opaque or back in under a minute and do not degrade over time. The prototypes are plates of conductive glass outlined with metal ions that spread out over the surface, blocking light, in response to electrical current.

Dynamic windows have the potential to cut heating and cooling costs or the need for blinds, but the technology is yet to catch on in the marketplace. Smart windows already being sold, such as those used on airlines, are made of materials such as tungsten oxide that change colour when charged with electricity. But these materials tend to be expensive, have a blue tint, can take over 20 minutes to



This photograph shows a smart window prototype dimming in response to electricity. This window is notable for possessing neutral colour, high contrast and excellent durability (Image courtesy: Yue et al./Joule 2017)

dim, and become less opaque over time.

"We did not tweak what was out there, we came up with a completely different solution," said senior author Michael McGehee, a Stanford University professor of materials science and engineering with a background in solar cells. McGehee and his group's prototype blocks light through the movement of copper and another metal in a solution over a sheet of transparent indium tin oxide modified by platinum nanoparticles. When transparent, the windows are clear and allow about 80 per cent of surrounding natural light through, and when dark, transmission drops to under 5 per cent. The researchers switched the windows on and off at least 5500 times and saw no change in the transmission of light, indicating that the design is durable.

Sunglasses generate solar power

Researchers from Karlsruhe Institute of Technology (KIT) have developed sunglasses with coloured, semitransparent



These solar glasses with lens-fitted semitransparent organic solar cells supply two sensors and electronics in the temples with electric power (Image courtesy: KIT)

solar cells applied onto lenses that supply a microprocessor and two displays with electric power. This paves the way for future applications such as the integration of organic solar cells into windows or overhead glazing.

The 'smart' solar glasses, designed by Dr Alexander Colmann and his team at KIT, are self-powered to measure and display the solar illumination intensity and ambient temperature. Solar cell lenses, perfectly fitted to a commercial frame, have a thickness of approx. 1.6 millimetres (mm) and weigh about six grams—just like the lenses of traditional sunglasses. The microprocessor and the two small displays are integrated into the temples of solar glasses. These show the illumination intensity and the ambient temperature as bar graphs.

Solar glasses also work in indoor environments under illumination down to 500 Lux, which is the usual illumination of an office or a living area. Under these conditions, each of the smart lenses still generates 200 microwatts of electric power—enough to operate devices such as a hearing aid or a step counter.